Optimal Time-Stepping with Non-Iterative Defect Correction for Coupled Reaction–Diffusion Equations

Aditya Tiwari

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Repository: `OTS-NIDC` (this paper is generated from it)

1 Review for Single Variable PDEs

Throughout, $\delta_x^2 u_j = (u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$ is the centered second difference and refinement means $\Delta x \to 0$ with Δt slaved to Δx.

1.1 The scalar family and the optimal time step

Consider the deterministic family

$$u_t = D\,u_{xx} + \alpha\,u, \qquad D > 0,\ \alpha \in \mathbb{R}, \tag{1}$$

whose stochastic motivation (the moment equations of (**??**), with $\alpha = \frac{k(k-1)}{2}\sigma^2$) is deferred to §**??**. The pure diffusion case $\alpha = 0$ is exactly Chu's setting (Chu, 2011).

Apply forward Euler in time and the three-point stencil in space. Taylor expansion of both sides, followed by repeated differentiation of (1) to convert time derivatives of u into space derivatives (the operators $D\partial_{xx}$ and α commute, giving the binomial identity $u_{t^k} = (D\partial_{xx} + \alpha)^k u$), yields the local truncation error organized by derivative type:

$$\tau \;=\; \underbrace{\left(\tfrac{D^2\Delta t}{2} - \tfrac{D\Delta x^2}{12}\right)u_{xxxx}}_{\text{Group 1}} \;+\; \underbrace{\alpha D\,\Delta t\,u_{xx}}_{\text{Group 2}} \;+\; \underbrace{\tfrac{\alpha^2\Delta t}{2}\,u}_{\text{Group 3}} \;+\; O(\Delta t^2, \Delta x^4). \tag{2}$$

Only Group 1 couples Δt to Δx. Setting it to zero:

Theorem 1 (Optimal time step (Chu, 2011))**.** *For (1) discretized by forward Euler / centered differences, the unique step size annihilating the leading (u_{xxxx}) truncation term is*

$$\boxed{\Delta t_{\text{opt}} = \frac{\Delta x^2}{6D}} \tag{3}$$

and no single step annihilates both the u_{xxxx} and u_{xxxxxx} truncation terms, since the level-2 condition is $\Delta t = \Delta x^2/(\sqrt{60}\,D) \neq \Delta t_{\text{opt}}$.

(Proof: one-line cancellation in (2); uniqueness by the level-2 computation. Full algebra in `sign-change.tex` §1.)

Proposition 2 (α-independence)**.** *Δt_{opt} is independent of the reaction parameter α: the cancellation operates on the diffusive group alone. For the stochastic hierarchy this makes Δt_{opt} independent of the noise amplitude σ and of the moment order k.*

Following the derivation inspecting the truncation error, at $\Delta t = \Delta t_{\text{opt}}$ the Group-1 term vanishes and the surviving residual is

$$\tau\big|_{\Delta t_{\text{opt}}} = \frac{\alpha \Delta x^2}{6} u_{xx} + \frac{\alpha^2 \Delta x^2}{12 D} u + O(\Delta x^4). \tag{4}$$

Then the truncation error τ as assembled in (2) is the *forward* residual: exact − scheme $= \Delta t\, \tau + \cdots$. Cancelling it therefore requires *adding* $\Delta t\, \tau$.

Lemma 3 (Closed form correction term)**.** *The corrected factor at any given time step is,*

$$C_j^{(1)} = +\frac{\alpha\, \Delta x^4}{36 D} \delta_x^2 u_j^n + \frac{\alpha^2\, \Delta x^4}{72 D^2} u_j^n. \tag{5}$$

Theorem 4 (First-level accuracy)**.** *The corrected scheme*

$$u_j^{n+1} = u_j^n + \Delta t_{\text{opt}} \left[D\, \delta_x^2 u_j^n + \alpha\, u_j^n \right] + C_j^{(1)} \tag{6}$$

has step error $O(\Delta x^6)$ and global order 4: the correction $C^{(1)}$ is itself $O(\Delta x^4)$, so its evaluation with the second-order operator δ_x^2 contributes only $O(\Delta x^6)$ to the step — the three-point stencil is mathematically sufficient at $p = 1$.

Status: [V]. Measured orders and errors, with the harness described in Appendix **??**:

Table 1: Scalar $p = 1$ convergence, $u_t = D u_{xx} + \alpha u$ with $D = 0.1$, $\alpha\sigma^2 = 0.25$, Neumann BCs, $T = 0.05$; L^2 error against the exact cosine-series solution. Data: `code/implementation/results.md` (generated by `convergence_test.py`, 2026-07). All methods use the same grid and step size Δt_{opt} at each level.

Method	$N = 32$	$N = 64$	$N = 128$	$N = 256$
OTS-NIDC ($p = 1$)	8.54e-09	5.26e-10	3.27e-11	2.04e-12
Forward Euler	1.23e-05	3.02e-06	7.52e-07	1.88e-07
4th-order FD + FE	1.52e-05	3.79e-06	9.45e-07	2.36e-07
Reference expm	2.77e-05	6.81e-06	1.70e-06	4.23e-07
Measured order (log–log slope, mean over levels):				
OTS-NIDC	4.01	(FE / 4th-order FD / expm: 2.00–2.01)		

The 4th-order-FD row carries the paper's cleanest negative control: high-order spatial accuracy alone buys nothing when the step size is slaved to Δx^2 — the $O(\Delta t)$ time error caps the whole scheme at second order. The boost must come from the time-step structure; that is the method's thesis, verified.

The p-level tower and the stencil bound

move this into an appendix, perhaps?

The general correction is generated by powers of the operator, with the sign of Remark **??**:

$$C_j^{(\le p)} = +\sum_{\ell=1}^{p} \frac{\Delta t_{\text{opt}}^{\ell}}{(\ell+1)!} \big(D\mathcal{D}_2 + \alpha\big)^{\ell} u_j^n, \tag{7}$$

where $\mathcal{D}_2$ is a spatial operator whose accuracy must be raised as ℓ grows. Spatial discretization error breaks the binomial symmetry of the time expansion, giving the stencil-scaling bound:

Proposition 5 (Stencil bound; sketch)**.** *Global order $2p+2$ requires corrections evaluated on at least a $(2p+3)$-point stencil: $p=1$ needs the 3-point main stencil plus a 5-point evaluation inside $C^{(1)}$ when full fourth-order spatial accuracy is demanded of the correction itself; $p=2$ needs a 7-point stencil to represent the uncancelled u_{xxxxxx} residual.*

Status: [O]for $p \geq 2$ numerics (no harness yet; the bound's derivation is a consistency argument in `OTS-NIDC.md` §6). The $p=1$ row is [V]via Table 1.

1.2 Stability of Single Variable Systems

phrase all of this as lemmas please god

1.2.1 Setting and criterion

Set $u_j^n = G^n e^{ikj\Delta x}$. The three-point Laplacian has symbol

$$\delta_x^2 u_j^n \;\to\; -\frac{4}{\Delta x^2}\, s\, u_j^n, \qquad s = \sin^2(k\Delta x/2) \in [0,1].$$

Let $\mu = D\Delta t/\Delta x^2$, $\eta = \alpha\Delta x^2/D$ (α already contains σ^2 for the moment equations), and $a = \alpha\Delta t = \mu\eta$. The exact one-step factor is $E(s) = e^{-4\mu s + a}$, so $|E| \leq e^a$: for $\alpha > 0$ the equation grows, and $|G| \leq 1$ is the wrong requirement. We use the von Neumann condition with growth allowance,

$$|G(s)| \leq 1 + C\,\Delta t \quad \text{uniformly in } s \in [0,1], \tag{8}$$

along refinement paths $\Delta t = \mu\Delta x^2/D$ with μ fixed (Lax–Richtmyer). The sharper, natural comparison is $|G(s)| \leq e^a$.

1.2.2 Amplification factors

We frame OTS-NIDC as a perturbation of the simple Finite-Euler method, so we begin by deriving the amplification factor G_{FE} and then transform it into G_{NIDC} as follows in,

$$G_{\text{FE}} = 1 - 4\mu s + \mu\eta, \tag{9}$$

$$G_{\text{NIDC}} = 1 - 4\mu s + \mu\eta - \tfrac{\eta}{9}s + \tfrac{\eta^2}{72}, \tag{10}$$

the latter from $C^{(1)}$ in (5): $\frac{\alpha\Delta x^4}{36D}\cdot(-\frac{4s}{\Delta x^2}) = -\frac{\eta}{9}s$ and $\frac{\alpha^2\Delta x^4}{72D^2} = \frac{\eta^2}{72}$. Both are affine in s with negative slope, so the extremes over $s \in [0,1]$ are at $s=0$ (physical mode) and $s=1$ (grid mode).

1.2.3 Stability at the optimal step

At $\Delta t = \Delta t_{\text{opt}}$, $\mu = 1/6$, $\eta = 6a$, and

$$G_{\text{NIDC}}(s) = \left(1 - \tfrac{2}{3}s\right)(1+a) + \tfrac{a^2}{2}. \tag{11}$$

Compare $E(s) = e^{-2s/3}e^a$: the correction supplies exactly the $a^2/2$ term of e^a at $s=0$, and $-\frac{2}{3}as$ is the cross term. Since $1 - \frac{2}{3}s \in [\frac{1}{3}, 1]$,

$$\tfrac{1}{3}(1+a) + \tfrac{a^2}{2} \;\leq\; G_{\text{NIDC}}(s) \;\leq\; 1 + a + \tfrac{a^2}{2} \;\leq\; e^a \qquad (a \geq 0).$$

Hence $G_{\text{NIDC}} > 0$ for all s (no oscillatory modes) and $|G_{\text{NIDC}}| \leq e^{\alpha\Delta t}$: (8) holds for every $\eta \geq 0$, with no restriction on η. The offset $+\eta^2/72$ is therefore not destabilizing; it is what makes $G(0)$ match e^a to $O(a^2)$. For plain FE at the same μ, $G_{\text{FE}}(s) = (1 - \frac{2}{3}s) + a$, also inside $[\,\frac{1}{3} + a,\, 1 + a\,]$.

1.2.4 Grid-scale boundary for general μ

Requiring the lower side of (8) at $s = 1$, i.e. $G(1) \geq -(1 + \mu\eta)$:

$$\mu_{\mathrm{FE,max}} = \frac{2}{4 - 2\eta} = \frac{1}{2 - \eta}, \tag{12}$$

$$\mu_{\mathrm{NIDC,max}} = \frac{2 - \eta/9 + \eta^2/72}{4 - 2\eta}, \tag{13}$$

with ratio

$$\frac{\mu_{\mathrm{NIDC,max}}}{\mu_{\mathrm{FE,max}}} = 1 - \frac{\eta}{18} + \frac{\eta^2}{144}. \tag{14}$$

(The stricter test $G(1) \geq -1$ gives denominators $4-\eta$ and the same ratio.) At $\eta = 0.30$: $\mu_{\mathrm{FE,max}} = 0.5882$, $\mu_{\mathrm{NIDC,max}} = 0.5788$, ratio 0.98396 ($-1.6\%$). The shrink is $O(\Delta x^2)$, and the effective NIDC CFL is $\Delta t \lesssim (1 - \eta/18)\,\Delta x^2/(2D)$.

1.2.5 Scope of the correction ($\Delta t \neq \Delta t_{\mathrm{opt}}$)

$C^{(1)}$ is derived at $\Delta t = \Delta t_{\mathrm{opt}}$ and is independent of Δt. At $s = 0$,

$$G_{\mathrm{NIDC}}(0) - e^a \;\approx\; \frac{\eta^2}{72}\big(1 - 36\mu^2\big),$$

which is ≤ 0 iff $\mu \geq 1/6$. For $\mu < 1/6$ the excess is Δt-independent, so at fixed Δx with $\Delta t \to 0$ the accumulated growth $(1+\mathrm{const})^{T/\Delta t}$ diverges. Stability statements therefore refer to refinement paths μ fixed, where the excess is $O(\Delta t^2)$ and (8) holds; the $\Delta t/\Delta t_{\mathrm{opt}} \in \{0.5, \ldots, 2\}$ table is a sweep over such paths, not a fixed-Δx study. The method is designed for $\mu = 1/6$.

1.2.6 Pair equations

Replace $D \to D_i + D_j$, $\alpha \to \alpha_{ij}$; (11) holds verbatim with $\mu_{ij} = 1/6$ at $\Delta t_{\mathrm{opt}}^{(ij)}$. For the coupled discrete system there is still no independent proof (open item **??**).

Theorem 6 (Von Neumann stability)**.** *With $s = \sin^2(k\Delta x/2)$, $\mu = D\Delta t/\Delta x^2$, $\eta = \alpha\Delta x^2/D$, the amplification factors are $G_{\mathrm{FE}} = 1 - 4\mu s + \mu\eta$ and*

$$G_{\mathrm{NIDC}} = 1 - 4\mu s + \mu\eta - \tfrac{\eta}{9}s + \tfrac{\eta^2}{72}.$$

For the moment hierarchy $\eta \geq 0$ and the grid-scale stability bound tightens from $\mu \leq 2/(4-\eta)$ to $\mu \leq (2 - \eta/9 + \eta^2/72)/(4-\eta)$: a relative shrink of $1 - \eta/18 + O(\eta^2)$, i.e. $O(\Delta x^2)$, about -1.6% at the coarse $\eta = 0.3$ test point. The optimal step $\mu = 1/6$ sits a factor of 3 inside the CFL ceiling for both schemes.

Status: [V](analytic derivation above + numerical check in `notebooks/ai-src/stability/`; the earlier "expands the CFL" conclusion was an artifact of the pre-correction sign and has been fixed in the same release). The correction is stability-neutral to first order; it never threatens the optimal step. Note the honest asymmetry: this is a *scalar* analysis; the coupled discrete schemes of §2 have no independent stability proof (open item **??**).

1.3 Efficiency profile

The method's cost argument is architectural, not just arithmetic: versus multi-stage Runge–Kutta, OTS-NIDC is single-pass (one read, one write of the state vector; no stage storage, no stage barriers); versus compact schemes, it is strictly local (no tridiagonal solves), so it maps onto GPUs and bandwidth-bound hardware without global communication. At $N = 256$ the verified scalar runs show $\sim 10^3$–$10^4\times$ lower error than forward Euler at identical grid and step [V]— order wins over constants at refinement.

2 The Step to Coupled PDEs

We now couple species to allow for multi-linear systems, a direct expansion from the canon work we expand on!

2.1 Reaction-Diffusion Family and OTS

For N species with diagonal diffusion, linear reaction coupling, and (here) zero noise — the deterministic skeleton of the moment systems to come —

$$\partial_t u^{(i)} = D_i\, \partial_{xx} u^{(i)} + \sum_j R_{ij} u^{(j)}. \tag{15}$$

Lemma 7 (OTS-NIDC for fixed indices)**.** *For equation (15), the optimal step for each $u^{(i)}$ individually, as derived in the single variable case, is exactly $\Delta t^{(i)}_{\text{opt}} = \frac{\Delta x^2}{6D_i}$, with first-level corrections $C^{(1),(i)}$ of the same algebraic form as (5), derived term-by-term in `OTS-NIDC.md` §8–9.*

With this, we get the desired theorem as follows.

Theorem 8 (First-level accuracy for fixed indices)**.** *The scheme*

$$u_k^{(i),n+1} = u_k^{(i),n} + \Delta t_{\text{opt}} \left[D_i \delta_x^2 u_k^{(i),n} + \sum_j R_{ij} u_k^{(j),n} \right] + C^{(1),(i)} \tag{16}$$

has similar step error and global order as given by Theorem 4.

2.1.1 Competing Time Steps and Subcycling

Notice that the scheme presented by Theorem 8 relies on having $u_k^{(j),n}$ from having the value of separate PDE $u^{(j)}$ at time-step n. However, since each equation itself relies on their own time-step, we lose the guarantee of having the required value for any given equation readily available if $D_i \neq D, \forall i$ such that $\Delta t^{(i)}_{\text{opt}} = \Delta t_{\text{opt}}$ for the whole system.

The two failure modes of a naive single step frame the problem.

Failure 1: shared step without correction. If species 1 (fast, D_1) is forced onto species 2's optimum $\Delta t^* = \tau_2$, its Group-1 cancellation breaks. The one-step identity

$$e(\Delta t^*) \;=\; \Delta t^* D_1\, \tilde{u}_{xxxx} \left[\tfrac{\Delta t^* D_1}{2} - \tfrac{\Delta x^2}{12} \right] + O((\Delta t^*)^3) \tag{17}$$

is exact for every Δt^*, and writing $\Delta t^* = \tau_1 + \epsilon$ turns the bracket into $\epsilon D_1/2$ *identically* (no Taylor truncation in ϵ). Hence:

Theorem 9 (Exact mismatch correction)**.** *For any shared step $\Delta t^* = \tau_1 + \epsilon > 0$ that is stable for species 1, subtracting (17) (with $\tilde{u}_{xxxx}$ from a 5-point stencil) restores species 1's full accuracy. The correction is algebraic, not asymptotic, in ϵ — including ϵ of the same size as τ_1 itself.*

Status: [V](worked example $D_1 = 3, D_2 = 1$ at $r = 3$, `examples/test_regime1_D1_3_D2_1.py`; correction there is grid-independent — the Δx^4 cancels, leaving pure stencil weights).

We remain, however, that this stencil is not ideal for each step, and it leads to questions of stability. In fact,

Failure 2: naive splitting. Lie–Trotter splitting with the coupling frozen over a macro step H incurs a sustained $O(H)$ forcing error; accumulated over T/H cycles this is a global $O(H) = O(\Delta x^2)$ error — worse than the uncorrected base scheme. Freezing is not viable; coupling data data must be interpolated.

Therefore, we introduce a subcycle protocol that fares to match timesteps where information can nicely be interpolated. First we find that,

Theorem 10 (Regime boundary)**.** *Forcing the fast species onto the slow species' optimal step $\Delta t^* = \tau_2 = \Delta x^2/6D_2$ is stable for the fast species iff*

$$r := D_1/D_2 \;\leq\; 3.$$

For $r > 3$ the shared step is unconditionally unstable and subcycling is mandatory.

Proof sketch. FTCS stability needs $\Delta t^* \leq \Delta x^2/2D_1$; substituting $\Delta t^* = \Delta x^2/6D_2$ gives $r \leq 3$. The threshold is grid-independent because both sides scale as Δx^2. □

Status: [S]— derived from plain FTCS; the corrected scheme's stability region (Prop. 6 is scalar-only) has not been re-derived, so the boundary is supported, not proven, for the corrected coupled scheme (open item **??**).

Then, via this regime boundary, we can strictly define the two regimes:

Regime 1: Single unoptimal step with exact mismatch correction. For any two equations with distinct $\Delta t^{(1)}_{\text{opt}} < \Delta t^{(2)}_{\text{opt}}$ and $r = D_2/D_1 \leq 3$, we evolve both equations by $\Delta t^*_{\text{opt}} = \Delta t^{(2)}_{\text{opt}}$ such that $u(2)$ is optimally solved by nature, and the discretization of $u^{(1)}$ is evolved with $\Delta t^{(1)}_{\text{opt}} + \epsilon$ such that the explicit mismatch correction term recovers order 4 accuracy.

Regime 2: Sub-cycling with Hermite Polynomial interpolation. For any two equations with distinct $\Delta t^{(1)}_{\text{opt}} < \Delta t^{(2)}_{\text{opt}}$ and $r = D_2/D_1 > 3$, evolve the fast equation $u^{(1)}$ via $n = \lfloor r \rfloor$ sub-cycled steps of $\Delta t^{(1)}_{\text{opt}}$ and a single remainder step $\rho = \Delta t^{(1)}_{\text{opt}} - n\Delta t^{(2)}_{\text{opt}}$ with explicit mismatch correction given by ϵ in $\rho = \Delta t^{(2)}_{\text{opt}} - \epsilon$. The slow equation $u^{(2)}$ can evolve with it's own $\Delta t^{(2)}_{\text{opt}}$, but requires cubic Hermite interpolation to overcome the $O(H)$ forcing error through $p = 2$ (Gear and Wells, 1984; Fok and Rosales, 2010)).

With these regimes, we complete the derivation of OTS-NIDC for coupled PDEs.

2.2 Stability

As we've established that the regime for individual equations remains the same, we prove stability by showing the correction factor described in Theorem 9 is similarly a perturbation on the simple FE scheme, and then analyzing the stability of the hermite interpolation separately.

We follow the scalar analysis of §1, now allowing the actual step h to differ from the optimal step $\tau = \frac{\Delta x^2}{6D}$. Write $h = \tau + \varepsilon$ and $r = \frac{h}{\tau}$. We first establish stability of the mismatch-corrected diffusion step, then restore the reaction terms.

Lemma 11 (Mismatch correction and amplification factor)**.** *For $u_t = Du_{xx}$, the corrected update is $u^{n+1} = u^n + hD\delta_x^2 u^n + \frac{hD^2\varepsilon}{2}\delta_x^4 u^n$, where $\delta_x^4 = (\delta_x^2)^2$. With $s = \sin^2\left(\frac{k\Delta x}{2}\right) \in [0,1]$, its amplification factor is $G_{\text{diff}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$.*

Proof. As in §1, the leading exact-minus-scheme residual is $hD\left(\frac{Dh}{2} - \frac{\Delta x^2}{12}\right)u_{xxxx}$. Since $h = \tau + \varepsilon$, the bracket equals $\frac{D\varepsilon}{2}$, so cancellation requires adding $\frac{hD^2\varepsilon}{2}\delta_x^4 u^n$.

The symbols of δ_x^2 and δ_x^4 are $-\frac{4s}{\Delta x^2}$ and $\frac{16s^2}{\Delta x^4}$. Substituting $h = r\tau$ and $\varepsilon = (r-1)\tau$ gives $G_{\text{diff}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$. At $r = 1$, the mismatch vanishes and this reduces to the optimal-step diffusion factor from §1. □

Unlike the reaction corrections in §1, the mismatch contribution need not vanish under refinement at fixed grid frequency: its symbol is $\frac{2r(r-1)}{9}s^2$. We therefore bound the complete factor rather than assume that the correction is uniformly small.

Lemma 12 (Diffusive stability interval)**.** *On an infinite or periodic uniform grid, the mismatch-corrected diffusion step satisfies $|G_{\text{diff}}(s)| \le 1$ for every $s \in [0,1]$ whenever $0 < r \le 4$.*

Proof. The upper bound follows from $G_{\text{diff}}(s) - 1 = \frac{2rs}{9}\left((r-1)s - 3\right) \le 0$, since $0 < r \le 4$ and $0 \le s \le 1$.

For the lower bound, completing the square gives $G_{\text{diff}}(s) = \frac{2}{9}\left(rs - \frac{s+3}{2}\right)^2 + 1 - \frac{(s+3)^2}{18} \ge \frac{1}{9}$. Hence $\frac{1}{9} \le G_{\text{diff}}(s) \le 1$.

For $r > 4$, the grid-scale factor is $G_{\text{diff}}(1) = 1 + \frac{2r(r-4)}{9} > 1$. Thus $r = 4$ is the sharp boundary for this Fourier criterion. □

In particular, the correction preserves diffusive stability throughout the imposed forward Euler range $0 < r \le 3$, as well as for shortened remainder steps $0 < r < 1$. We retain $r \le 3$ as the operating restriction and use it below to recover the reaction-growth bound from §1.

For $u_t = Du_{xx} + \alpha u$, the reaction correction must also be evaluated at the actual step. The same Taylor expansion used in §1 gives $C_{\text{react}}u^n = h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. At $h = \tau$, this is exactly the correction previously derived. Together with the mismatch term, it cancels the leading residual at a general step h.

Theorem 13 (Scalar stability with mismatch correction)**.** *Let $\alpha \ge 0$, and consider $u^{n+1} = u^n + h(D\delta_x^2 u^n + \alpha u^n) + \frac{hD^2(h-\tau)}{2}\delta_x^4 u^n + h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. On an infinite or periodic uniform grid, if $0 < h \le 3\tau$, then its amplification factor satisfies $|G(s)| \le e^{\alpha h}$ uniformly in $s \in [0,1]$.*

Consequently, any sequence of admitted steps, including optimal substeps and shortened remainder steps, satisfies $\|u^N\|_{\ell^2} \le e^{\alpha T}\|u^0\|_{\ell^2}$, where $T = \sum_{n=0}^{N-1} h_n$.

Proof. Set $a = \alpha h$ and $r = \frac{h}{\tau}$. Substituting the Fourier symbols gives $G(s) = G_{\text{diff}}(s) + a\left(1 - \frac{2r}{3}s\right) + \frac{a^2}{2}$. The preceding lemma gives $|G_{\text{diff}}(s)| \le 1$. Moreover, for $0 < r \le 3$, $-1 \le 1 - \frac{2r}{3}s \le 1$. Therefore, $|G(s)| \le 1 + a + \frac{a^2}{2} \le e^a$.

Over successive steps, the amplification factors multiply. Hence their product has magnitude at most $\prod_{n=0}^{N-1} e^{\alpha h_n} = e^{\alpha T}$. The discrete ℓ^2 estimate follows from Parseval's identity. □

Thus the mismatch correction preserves the scalar growth allowance used in §1 throughout the admitted forward Euler range. In a subcycle, full steps have $r = 1$ and any shortened remainder has $0 < r < 1$, so both satisfy the same estimate. This establishes the scalar building block; stability of off-diagonal reaction coupling and its interpolation still requires analysis of the complete coupled update.

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